

SOURABH DONGRE

DATA SCIENTIST || ML ENGINEER

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CAREER OBJECTIVE

A results-driven Data Scientist and ML Engineer with 3+ years experience in Machine Learning, Computer Vision, and Generative AI. Committed to developing innovative, data-driven solutions that optimize operations and enhance decision-making. Skilled in building and deploying AI models to address complex challenges in various industries. Seeking to contribute to an organization focused on leveraging cutting-edge AI technologies for business transformation

EXPERIENCE

● Datadelve Technologies Private Limited

DATA SCIENTIST

August 2021 - present

- **Customer Behavior Modeling:** Developed and optimized machine learning models using Python to predict customer behavior, enhancing decision-making and marketing strategies.
- **Computer Vision Implementation:** Applied deep learning techniques PyTorch for object detection, image classification, and segmentation, automating visual inspection processes.
- **Data Preprocessing & Feature Engineering:** Preprocessed large datasets and applied feature engineering techniques to improve the accuracy and performance of machine learning models.
- **AI Model Deployment:** Managed end-to-end AI model deployment, integrating real-time predictions with APIs and databases, while utilizing JIRA for project management and progress tracking.
- **Generative AI Development:** Built and trained GPT-based models to develop customized content generation solutions for marketing, enhancing customer engagement and interaction.

SKILL SET

- **Key Skills** : Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, SciPy, PyTorch, Keras, MySQL
- **Machine Learning** : Linear Regression, Logistic Regression, Decision Tree, Random Forest, XG boost, Naive Bayes, K-NN, Clustering-K-means and Hierarchical, SVM, NLP
- **Deep Learning** : YOLO, CNN, RNN
- **Generative AI** : GPT-3, Text Generation
- **Artificial Intelligence** : Generative AI, LLMs, LangChain
- **Tools** : GitHub, JIRA, Django, Azure

EDUCATION

- **Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal** 2021
 - M.Tech in Industrial engineering & management from JIT Borawan (CGPA : 7.77)
- **Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal** 2016
 - B.Tech in Mechanical engineering from JIT Borawan (CGPA : 7.77)

PROJECTS

● Project 1: AI Powered Human Detection in Warehouse using YOLOv8

Description:

Developed an AI-powered human detection system for warehouses using YOLOv8 utilizing high-definition cameras feed to detect human presence and movement in real-time within warehouse. This helped in improving warehouse security, monitor worker safety for efficiency analysis and ensuring compliance with safety protocols.

Roles & Responsibilities :

- Trained the model on appropriate model architecture (YOLOv8) using custom dataset to enhance detection accuracy.
- Annotated images and videos of warehouse environments incorporating different human actions and potential obstacles.
- Integrated the model with existing warehouse surveillance cameras to enable real-time analysis and monitoring of personnel activity.
- Developed pipeline to track real-time human movements and generate alerts for security breaches and safety violations.
- Evaluated the model performance in various warehouse settings like different lighting, environmental, and traffic conditions.
- Model deployment and ensuring its scaling across multiple camera feeds and areas with low latency.
- Refined the model's performance based on real-time feedback, thus addressing potential edge cases, and ensuring system reliability.

Key Skills & Technologies:

- | | |
|--------------|---------------------|
| • OpenCV | • Pytorch |
| • NumPy | • Sklearn |
| • Matplotlib | • CNN |
| • Keras | • Transfer Learning |
| • YOLOv8 | • Model Deployment |
| • Labelme | • AWS Cloud |

● Project 2 : Developed Customer Churn Models for Enhanced Retention Strategies

Description:

The objective of the project is to identify customers who are at risk of churning, enabling businesses to implement targeted retention strategies and reduce customer attrition.

Roles & Responsibilities :

- Performed feature engineering and preprocessing techniques such as scaling ,handling outliers and missing values.
- Trained and evaluated by splitting the data into training and testing sets.
- Fit the selected model on the training data and evaluate its performance on the testing data using metrics such as accuracy, precision, recall, and F1-score.
- Fine-tuned the hyperparameters of the model to optimize its performance.
- Interpreted the results to understand the importance of different features in predicting customer churn.
- Deployed the model in AWS Cloud.

Key Skills & Technologies:

- | | |
|-----------------|-----------------------|
| • Python | • Ensemble Models |
| • NumPy | • Logistic Regression |
| • Pandas | • Confusion matrix |
| • Sklearn | • Oracle Database |
| • Decision Tree | • Model Deployment |
| • Random Forest | • AWS Cloud |

● **Project 3: Educational Tutoring Chatbot**

Description:

The project aimed at building a chatbot to assist students, helping them with homework, explaining difficult concepts, and providing practice quizzes.

Roles & Responsibilities :

- Designed and developed the chatbot’s core structure, including conversational flows, user intent recognition, and responses for different educational topics.
- Integrated educational content from diverse sources (e.g., textbooks, websites, APIs) to ensure accurate and engaging material for students.
- Trained and fine-tuned the model to enable the chatbot to understand and respond to a wide range of student queries.
- Implemented adaptive learning algorithms to personalize the learning experience, adjusting content and difficulty based on student performance.
- Developed a dynamic quiz system with automated grading, hints, and explanations for correct/incorrect answers to reinforce learning.
- Continuously tested, deployed, and maintained the chatbot on web, ensuring scalability and regularly updating content based on user feedback.

Key Skills & Technologies:

- Python
- NumPy
- Pandas
- Sklearn
- Streamlit
- GPT-3.5
- FAISS
- Llama2

Algorithms:

- Generative AI
- LangChain
- LLMs (Language Learning Models)

● **Project 4 : Fraud Detection in Retail Transactions**

Description:

The project aim was to create an advanced AI-driven model to detect fraudulent retail transactions in real-time. The goal was to analyze vast amounts of transaction data to identify anomalies and prevent potential fraud before it could impact the business. This solution was aimed at enhancing the security of the retail system while maintaining a seamless user experience for legitimate customers.

Roles & Responsibilities :

- Preprocessed transaction data to identify features indicative of fraud, such as unusual patterns or location discrepancies.
- Developed classification models using Random Forest, XGBoost, and Isolation Forest for real-time fraud detection.
- Evaluated fraud detection models using confusion matrices, precision-recall, and AUC-ROC curves to ensure high accuracy.
- Deployed a real-time fraud alert system to flag suspicious activities for immediate intervention.
- Collaborated with retail and financial teams to enhance fraud detection models and reduce financial losses.

Key Skills & Technologies:

- Machine Learning
- Fraud Detection
- Anomaly Detection
- Python
- Data Preprocessing
- Random Forest
- XGBoost
- Isolation Forest
- Real-Time Anomaly Detection

DECLARATION

I hereby declare that the information and facts mentioned above are true the best of my knowledge.

SOURABH DONGRE